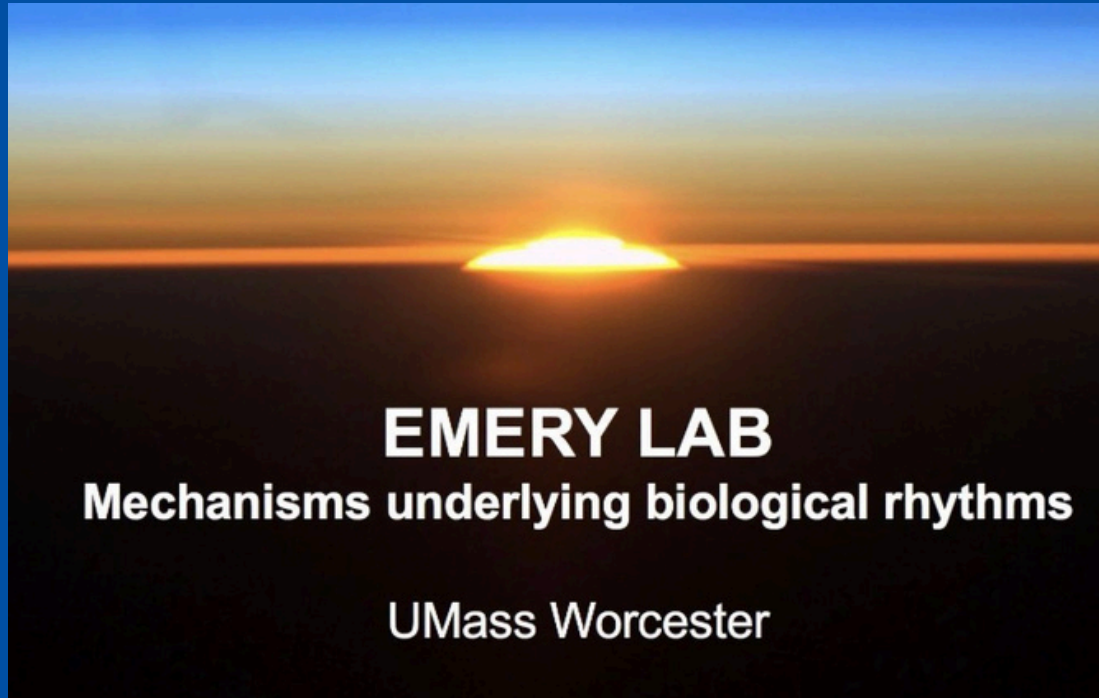


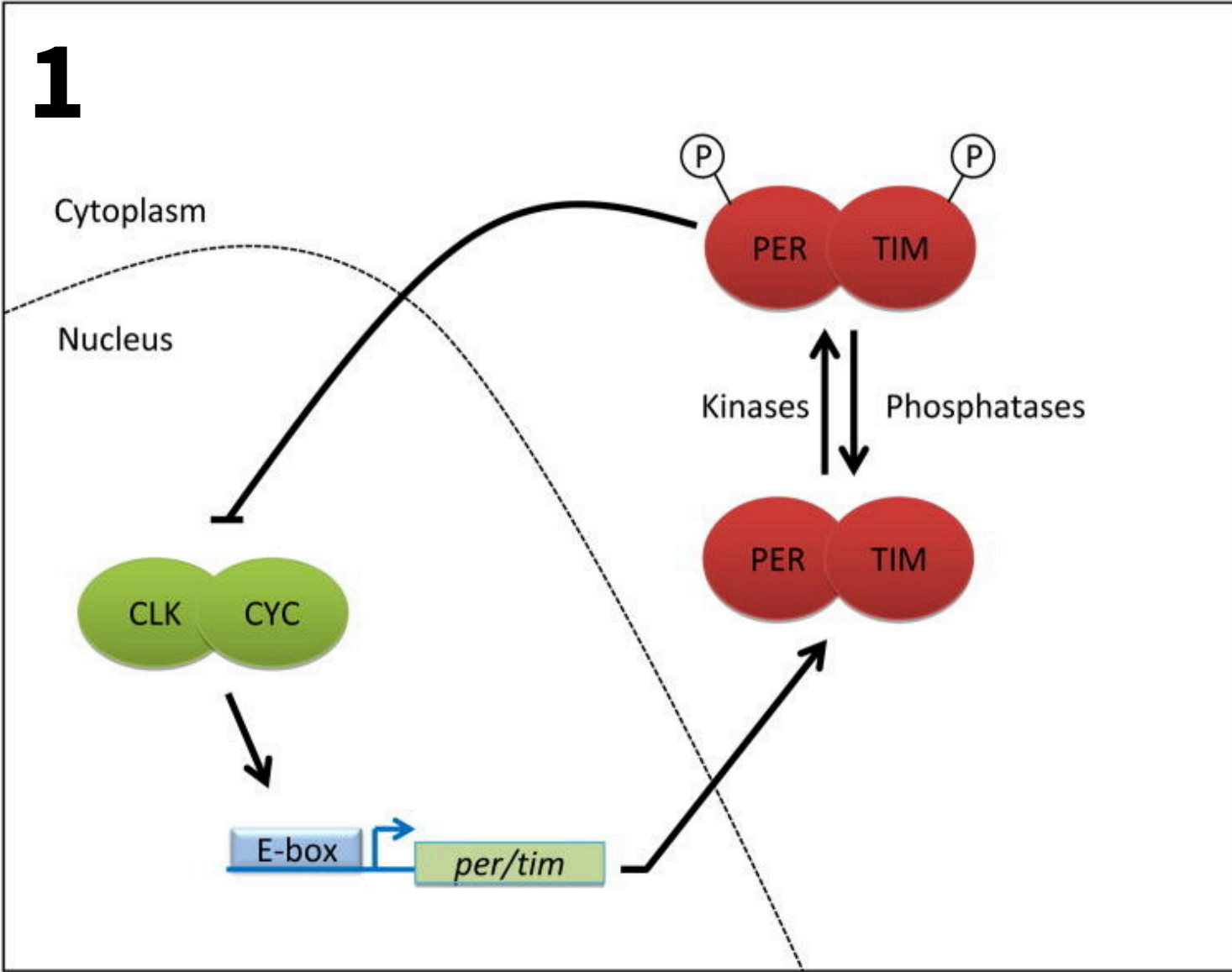
Evaluating the role of circadian clock genes in time-dependent behavioral and sleep responses to traumatic brain injury

Reya Satam, Olivia Heintz
Department of Neurobiology, UMass Chan Medical School



Introduction

Traumatic brain injury (TBI) produces persistent alterations in behavioral function, including disturbances in sleep and circadian regulation. The molecular clock in *Drosophila* consists of a negative feedback loop with four primary genes.



Previous work in our model identified time-of-day-dependent differences in post-TBI sleep phenotypes, suggesting that circadian biology influences TBI recovery. However, it remains unclear whether these temporal effects require a functional molecular clock. We investigate if disruption of clock function through gene dysregulation eliminates time dependent responses to TBI.

Hypotheses

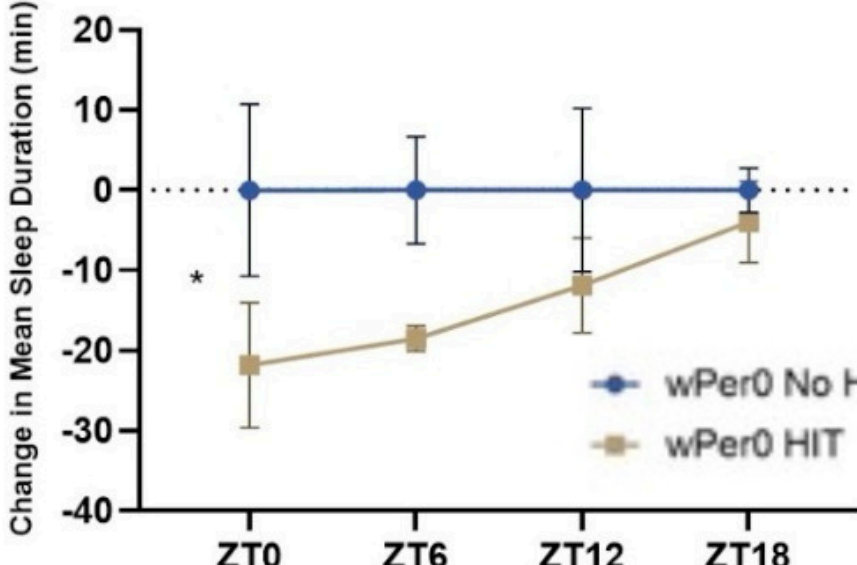
Disruption of molecular circadian clock function will reduce or eliminate time-of-day dependent differences in post-TBI sleep behavior in flies with varying genetic traits.

Disruption of exclusive neuronal clock function will reduce or eliminate time-dependent effects in post-TBI sleep behavior.

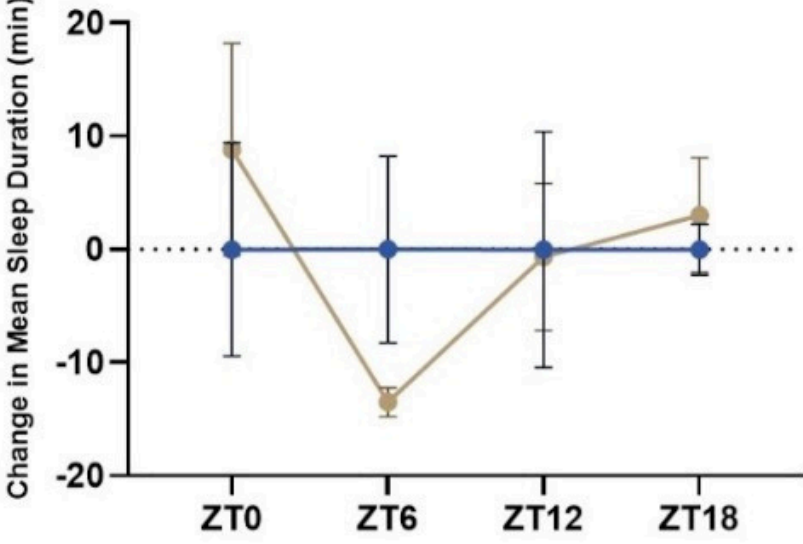
Results

Does PER disruption alter sleep effects in white-eyed flies?

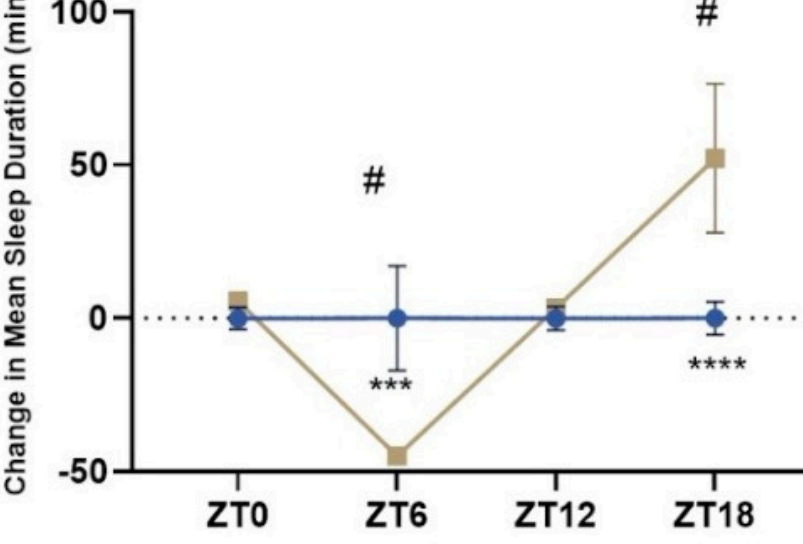
Daytime Male Mean Sleep Episode Duration



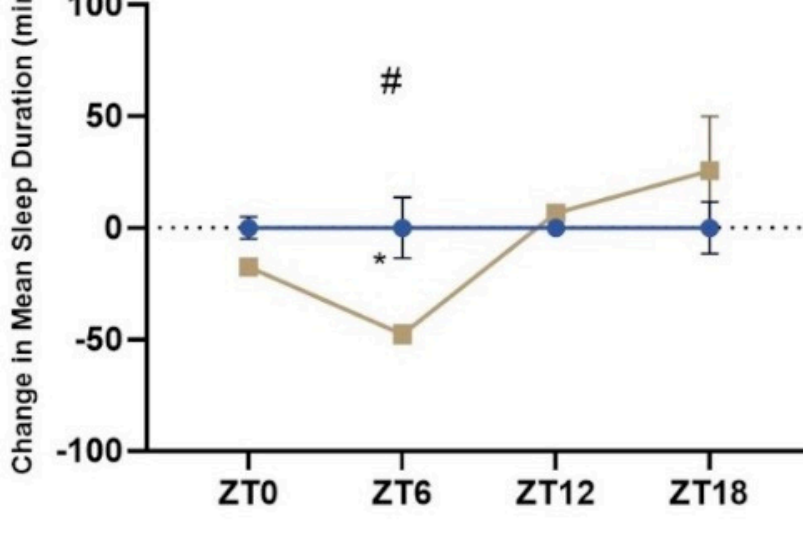
Daytime Female Mean Sleep Episode Duration



Nighttime Male Mean Sleep Episode Duration

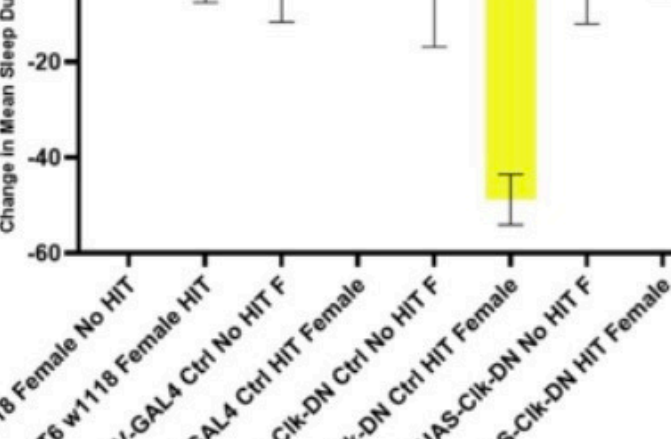
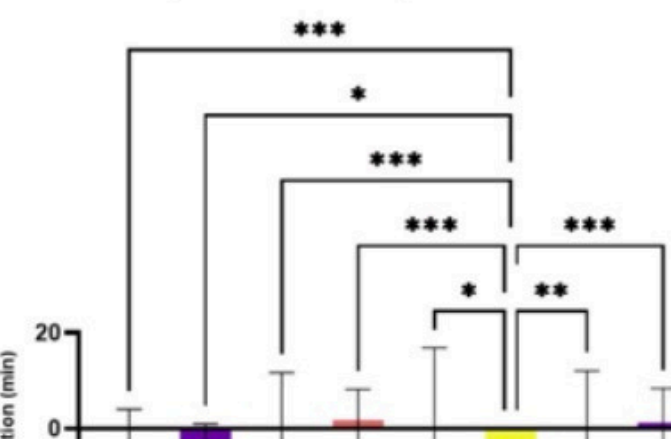
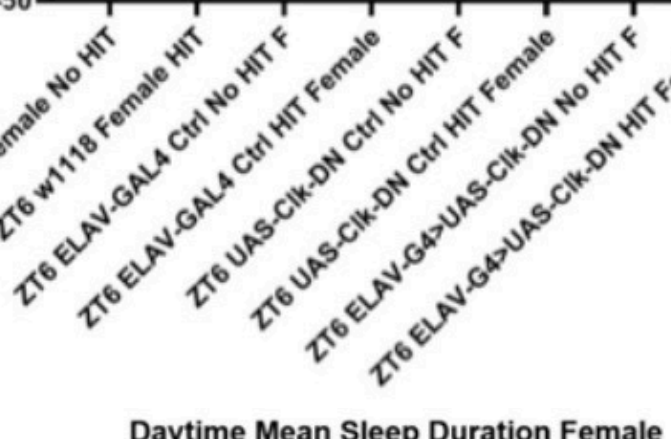
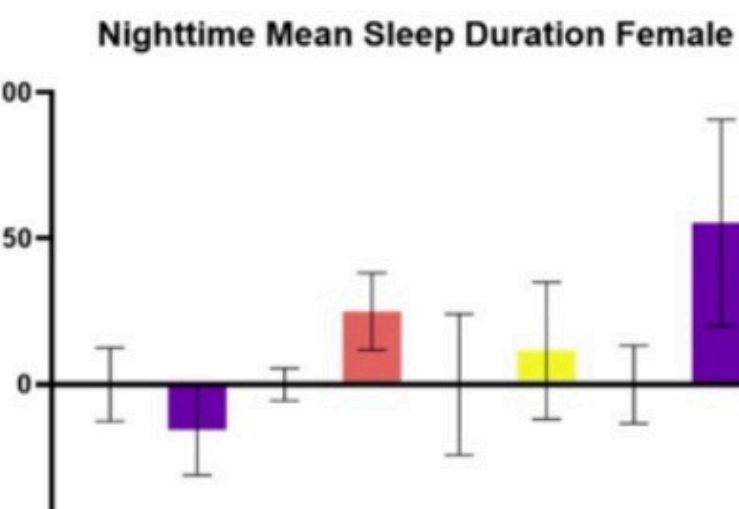


Nighttime Female Mean Sleep Episode Duration



wPER⁰ flies demonstrated partial time-dependent effects despite having a dysregulated clock. Mean sleep episode duration dipped for flies injured at ZT6, resembling wild-type time-dependent phenotype where ZT6 flies exhibit more fragmented sleep.

Does neuronal specific disruption of the clock eliminate the effects?



CLK-DN flies demonstrate no significant or consistent relative change in mean sleep episode to the wild-type flies, indicating that some aspects of the time-dependent effects persisted after dysregulation of the neuronal clock.

Discussion & Conclusions

The white-eyed flies with a disrupted molecular clock had only some time-related effects abolished compared to previously tested flies in a rosy background. This indicates that flies with varying eye genotypes could have different origins for time-dependent post-TBI reactions. White-eye mutations in *Drosophila* have demonstrated varied sleep-wake cycles and neurotransmitter signaling which indicates an additional neural pathway in the white-eyed flies could contribute to the effects.

Experimental flies with a disrupted molecular clock in neurons demonstrate little maintenance of time-related efforts, meaning neurons have little origin. Investigating the presence of other time-dependent effects in other brain cell types is a future direction for research.

Methodology

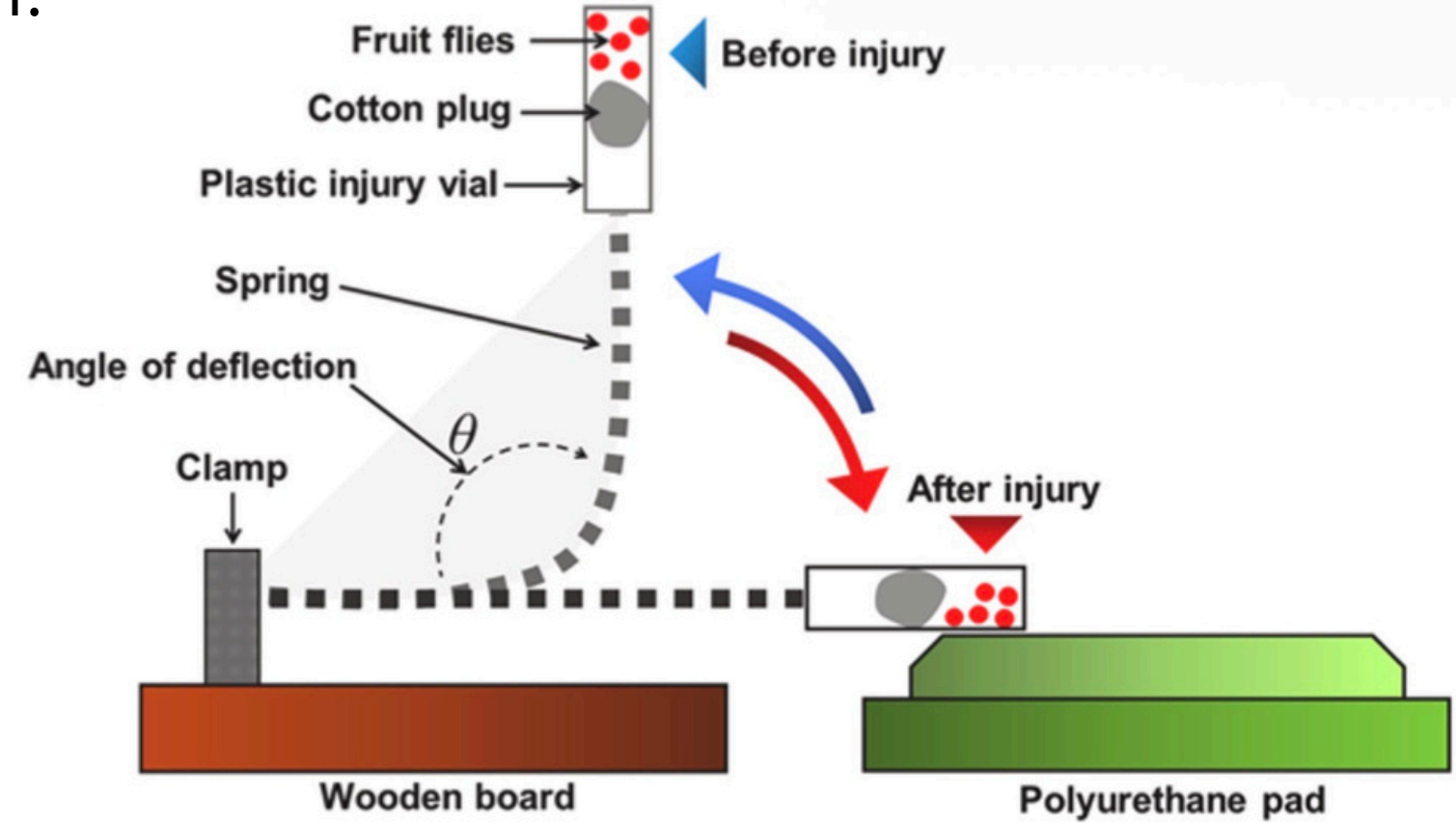
1: Circadian Manipulation

The experimental genotypes in this investigation include per⁰ and CLK-DN which was produced using the GAL4-UAS genetic driver system. The comparison groups were wild-type and sham-injured groups. Flies were grouped according to Zeitgeber time (ZT), where ZT0 = lights on and ZT12 = lights off.

2: TBI Paradigm

Experimental flies underwent diffuse TBI using a High Impact Trauma (HIT) device at designated ZT points. Following injury, flies were transferred to behavioral monitoring tubes for multi-day post-injury assessment.

2



3: Behavioral Recording

Locomotor activity of flies were continuously recorded for 5 days using infrared beam-break monitors provided by Drosophila Activity Monitors (DAM). Data were processed in MATLAB to quantify total sleep duration, mean sleep episode duration, and other metrics. Sleep architecture was evaluated to assess changes in post-TBI behavioral recovery.

4: Statistical Analysis

Behavioral outcomes were compared across genotype, injury status, status, and circadian time of impact using one-way and two-way ANOVA (normalization test and Tukey's post-hoc test). Data are presented as mean ± [SEM]. Statistical significance was defined as p < 0.05.

References

1. Tataroglu, and Emery. 2015. <https://pmc.ncbi.nlm.nih.gov/articles/PMC4049855/>
2. Aggarwal, et al. 2022. <https://pubmed.ncbi.nlm.nih.gov/35189165/>

Acknowledgements

Special thanks to Patrick Emery, all other members of the Emery lab, and Zoe Posner (M.D-PhD student at Yale)!